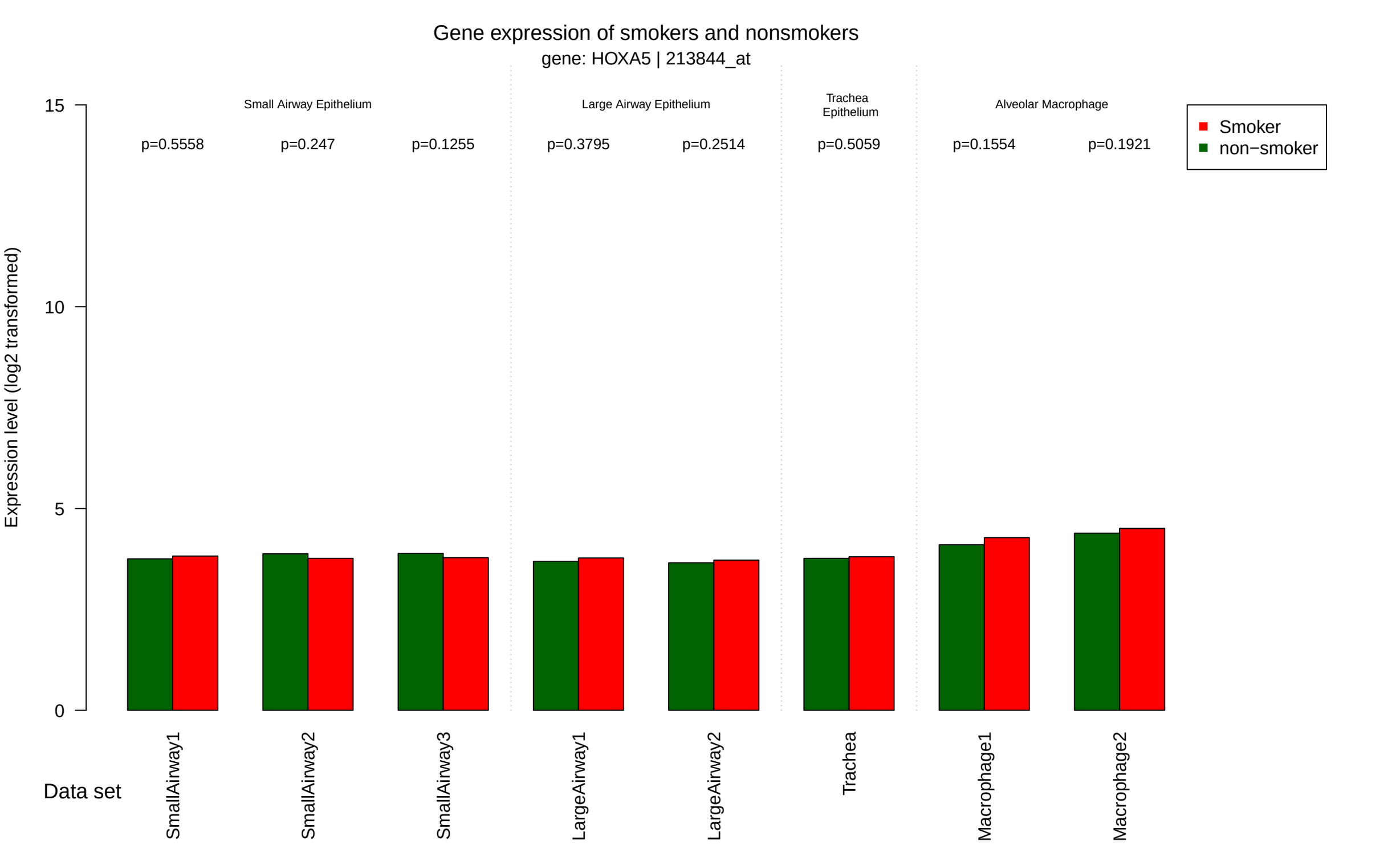
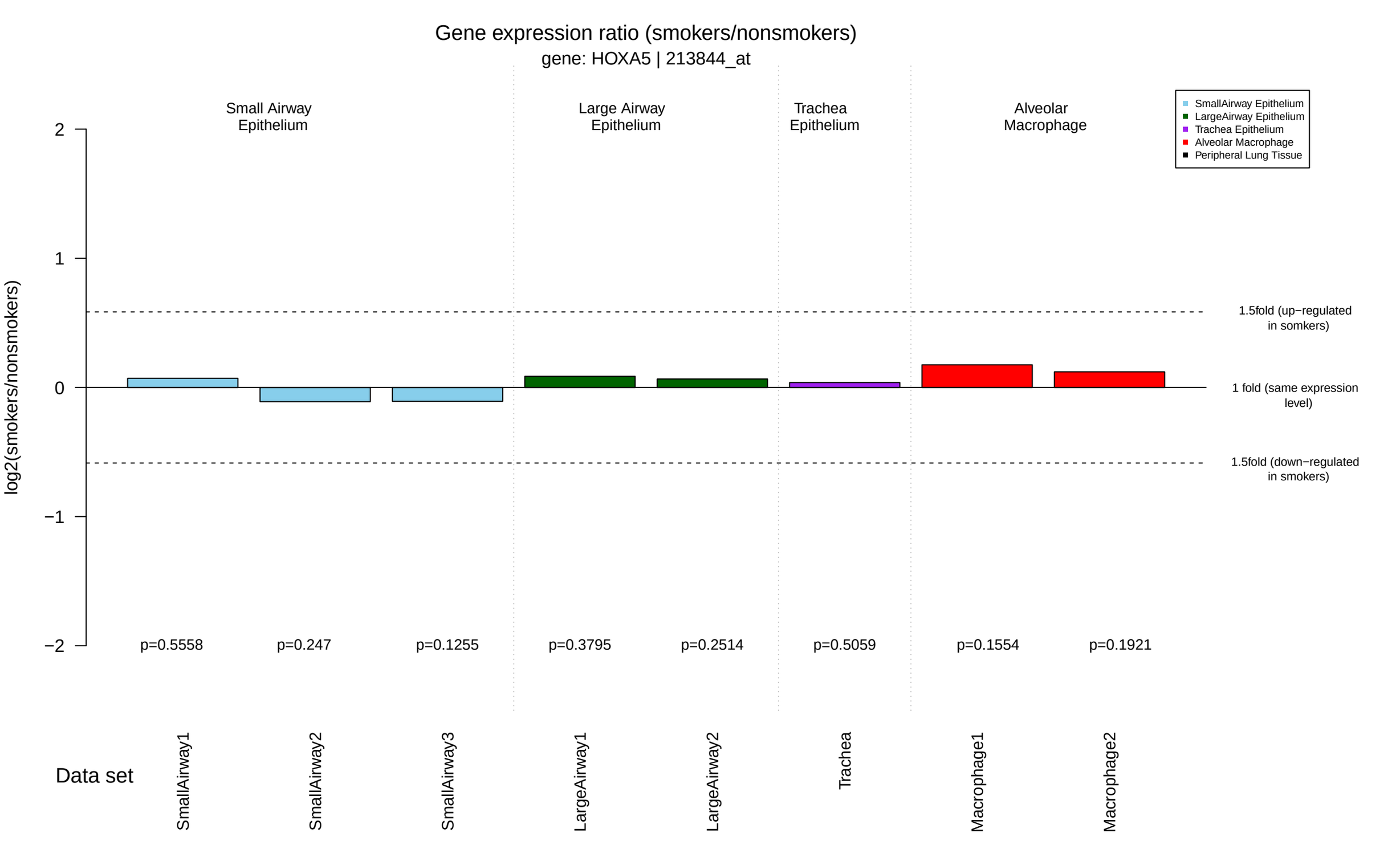
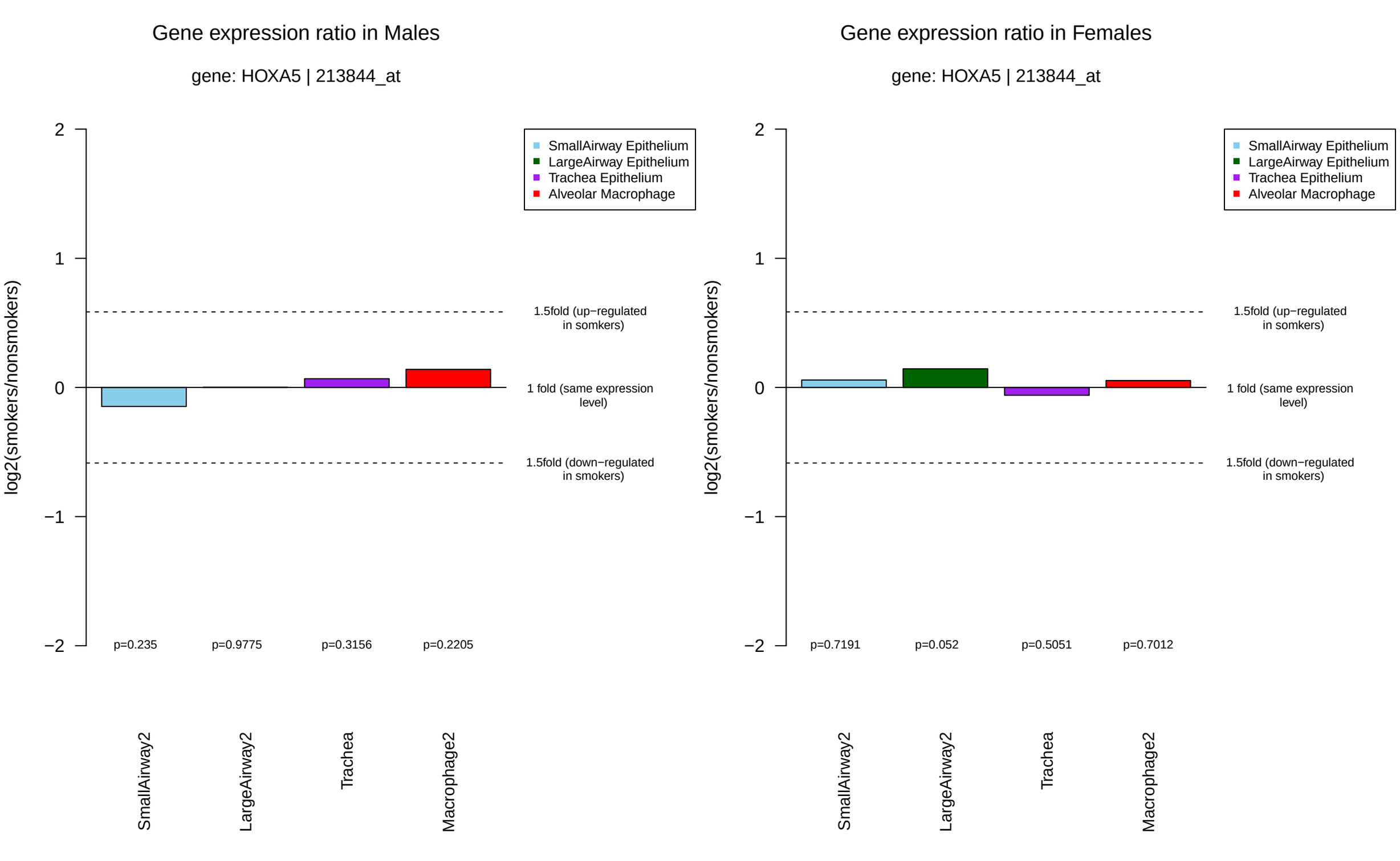
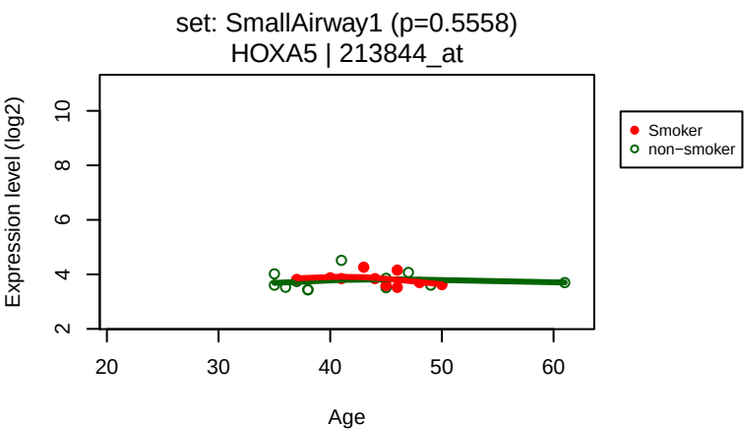
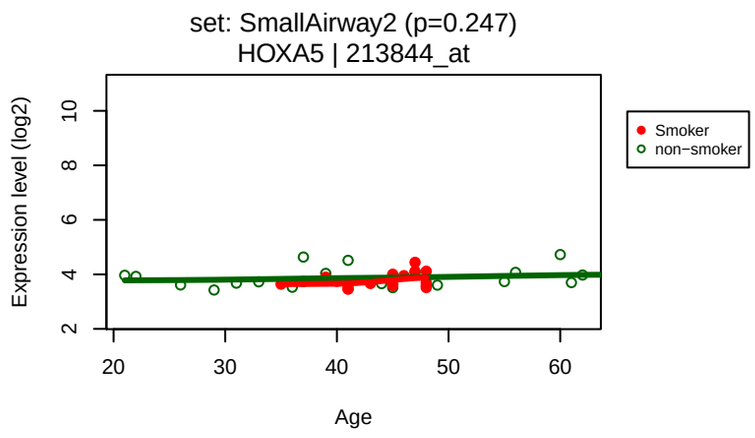
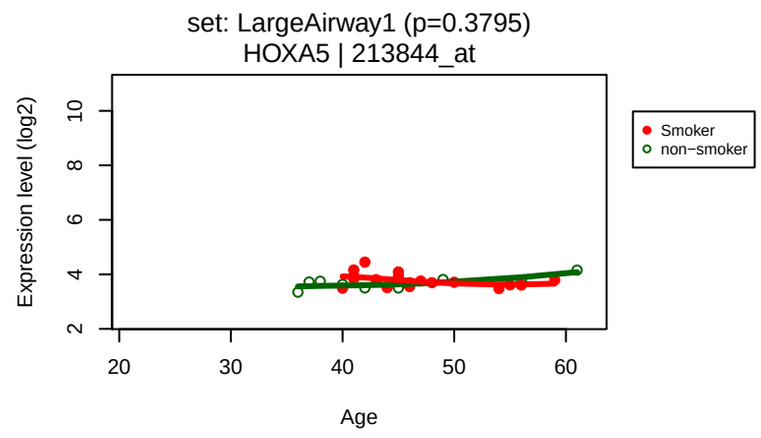

Gene expression of smokers and nonsmokers
gene: HOXA5 | 213844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5558
p=0.247
p=0.1255
p=0.3795
p=0.2514
p=0.5059
p=0.1554
p=0.1921
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

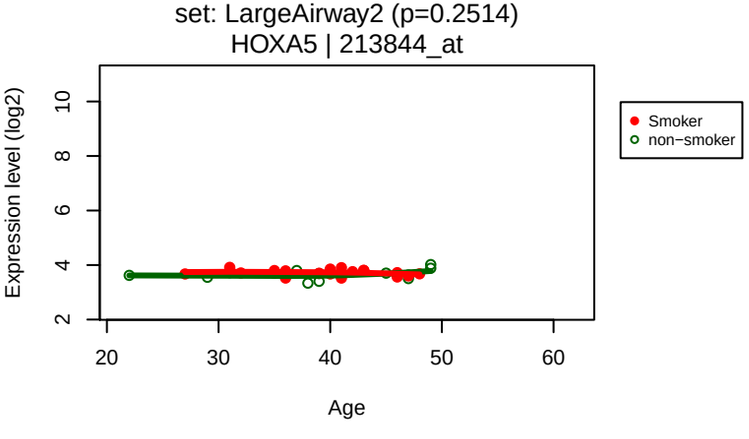

Gene expression ratio (smokers/nonsmokers)
gene: HOXA5 | 213844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5558
p=0.247
p=0.1255
p=0.3795
p=0.2514
p=0.5059
p=0.1554
p=0.1921
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

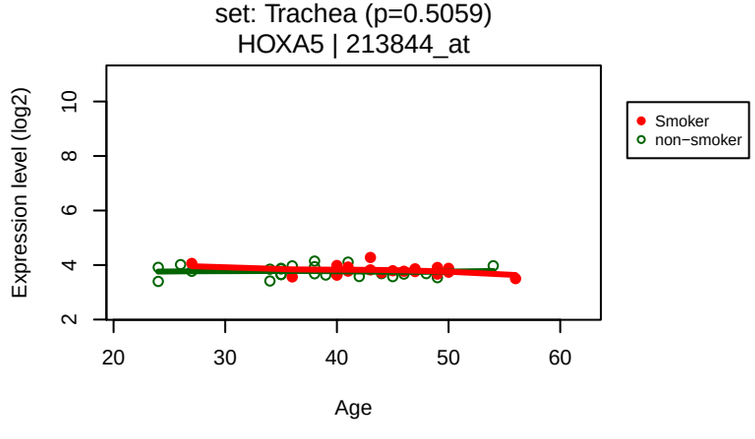

Gene expression ratio in Males
gene: HOXA5 | 213844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.235
p=0.9775
p=0.3156
p=0.2205
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXA5 | 213844_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7191
p=0.052
p=0.5051
p=0.7012
SmallAirway2
LargeAirway2
Trachea
Macrophage2

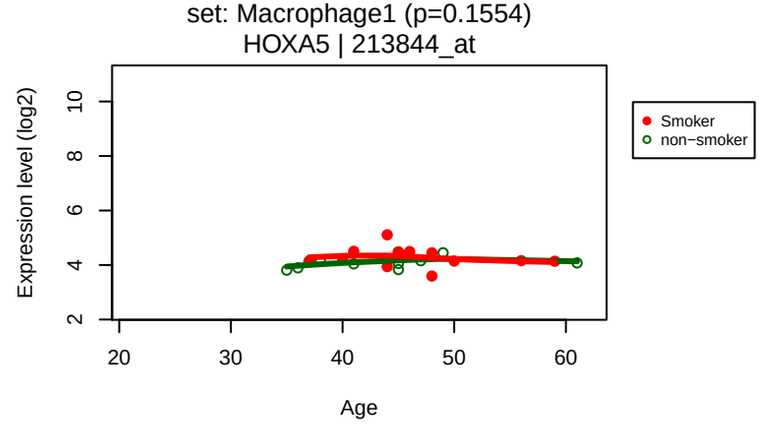

set: SmallAirway1 (p=0.5558)
HOXA5 | 213844_at
Expression level (log2)
Age
Smoker
non−smoker

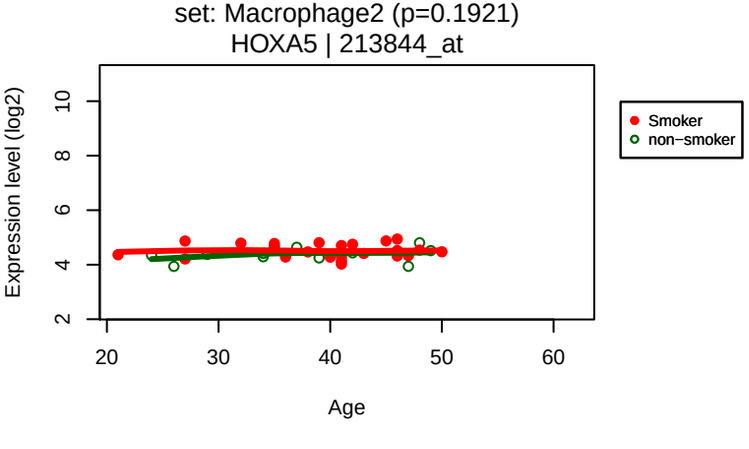

set: SmallAirway2 (p=0.247)
HOXA5 | 213844_at
Expression level (log2)
Age
Smoker
non−smoker

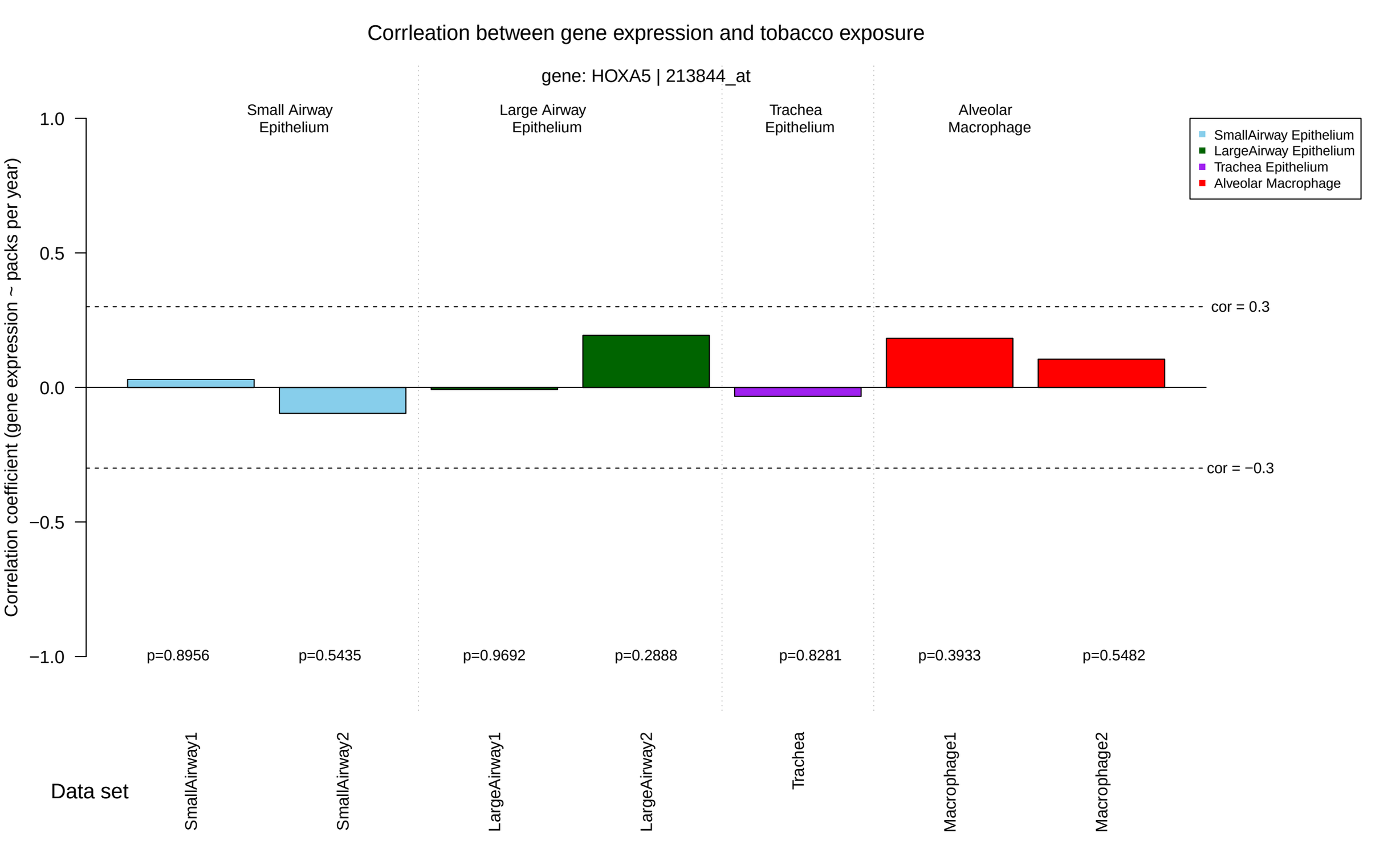

set: LargeAirway1 (p=0.3795)
HOXA5 | 213844_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2514)
HOXA5 | 213844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5059)
HOXA5 | 213844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1554)
HOXA5 | 213844_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1921)
HOXA5 | 213844_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXA5 | 213844_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8956
p=0.5435
p=0.9692
p=0.2888
p=0.8281
p=0.3933
p=0.5482
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2